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Bihar Engineering University, Patna

B.Tech 5th Semester Exam., 2020

Time: 3 hours

Code: 101502 *Subject: Environment Engineering-I* *Full Marks: 70*

Instructions:-

(i) The marks are indicated in the right-hand margin.
(ii) There are NINE questions in this paper.
(iii) Attempt FIVE questions in all.
(iv) Question No. 1 is compulsory.
(v) Assume data suitably, if not given.

Q.1 Choose the correct option of the following (any seven): **[14]**

(a) Normally designed period for a water supply scheme is
(i) up to 20 years (iii) 40-60 years
(ii) 20-40 years (iv) 60-80 years

(b) According to Indian Standards, water requirement per head per day in an office building is
(i) 15 litres (iii) 45 litres
(ii) 30 litres (iv) 60 litres

(c) Methemoglobinemia, the blue baby syndrome is caused by consuming water containing excess of
(i) fluoride (iii) nitrite
(ii) phosphate (iv) nitrate

(d) Colour of water is expressed in number of a
(i) po value (iii) platinum-cobalt scale
(ii) silica scale (iv) None of the above

(e) The maximum permissible chloride in water for domestic purpose should not exceed
(i) 250 mg/l (iii) 750 mg/l
(ii) 500 mg/l (iv) 1000 mg/l

(f) Suitable layout of water distribution system for a well-planned city is
(i) dead end system (iii) ring system
(ii) grid-iron system (iv) radial system

(g) Reflux valve is also known as
(i) sluice valve (iii) check valve
(ii) air valve (iv) drain valve

(h) Which of the following factors is primarily responsible for causing air pollution in modern days?
(i) Dust storms (iii) Forest fires
(ii) Industries and automobiles (iv) None of the above

(i) During temperature inversion in atmosphere, air pollutants tend to
(i) accumulate above inversion layer (iii) disperse laterally
(ii) accumulate below inversion layer (iv) disperse vertically

(j) As per IS: 4954-1964 an acceptable noise level for residential and business urban areas is
(i) 40-50 dB (iii) 15-25 dB
(ii) 30-40 dB (iv) 50-60 dB

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Q.2 Solve both questions : **[7+7=14]**

(a) What is the difference between BOD and COD? Calculate 2 days 30˚C BOD of sewage sample whose 5 days 20˚C BOD is 110 mg/l. Assume K_D at 20˚C as 0.1.

(b) Explain in detail the structure of atmosphere.

Q.3 Solve both questions : **[7+7=14]**

(a) Discuss the important characteristics of water used for domestic purpose. Draw the flow sheet of conventional treatment of water.

(b) What is the difference between disinfection and sterilization? Explain the different methods of disinfection.

Q.4 Solve both questions : **[7+7=14]**

(a) Define air pollution, and differentiate between natural and man-made air pollution.

(b) What is photochemical smog, how is it formed and how does it effect?

Q.5 Solve both questions : **[7+7=14]**

(a) Write a note on air pollution caused by automobiles and its control.

(b) What is meant by the effective height of a chimney and how is it computed?

Q.6 Solve both questions : **[7+7=14]**

(a) What is noise? What is its unit of measurement? What are the ill effects of noise pollution?

(b) Explain in brief the major factors and action that may help in noise abatement in society.

Q.7 Solve both questions : **[7+7=14]**

(a) Illustrate with sketches the different types of layouts of pipe systems in distributing water, and compare their merits and demerits.

(b) Discuss briefly the common stresses produced in pipelines used for conveying water.

Q.8 Solve both questions : **[7+7=14]**

(a) Briefly discuss the design principles involved in the design of a water supply network to be laid in multi-storeyed residential building.

(b) What factors will you keep in mind while designing plumbing system for water supply to a house?

Q.9 Write short notes on (any four): **[14]**

(a) Storage tank
(b) Superchlorination
(c) Sampling process
(d) Theory of coagulation process
(e) Reverse osmosis

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